Multi Insurance Creator

Course Name: GEN AI

Institution Name: Medicaps University – Datagami Skill Based Course

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1. Problem Statement

The insurance domain involves complex concepts such as risk assessment, premium calculation, and policy conditions, which are often difficult for users to understand through traditional text-based explanations.

Existing systems:

- Provide only textual responses
- Lack data-driven insights
- Do not generate visual explanations
- May produce inaccurate or hallucinated information

Problem:

There is a need for an intelligent system that can generate accurate, data-driven, and visually rich explanations of insurance concepts by combining multiple data sources and AI techniques.

2. Project Objectives:

The main objectives of this project are:

- To develop a multimodal AI system that generates both textual and visual outputs
- To integrate Retrieval-Augmented Generation (RAG) for accurate and context-aware responses
- To combine structured (datasets) and unstructured data (documents)
- To design an agent-based architecture for task specialization
- To generate professional infographics from user queries
- To reduce hallucinations and improve reliability of AI responses

3.Scope of the Project:

The scope of the project includes:

- Insurance-related query understanding
- Data analysis using structured datasets (e.g., driver age, accident history)
- Retrieval of policy-related knowledge using RAG
- Generation of explanations and visual infographics

Use in:

- Insurance education platforms
- Customer support systems
- Risk analysis tools

Limitations:

- Focused mainly on insurance domain
- Dependent on quality of datasets and documents
- Real-time deployment not fully covered

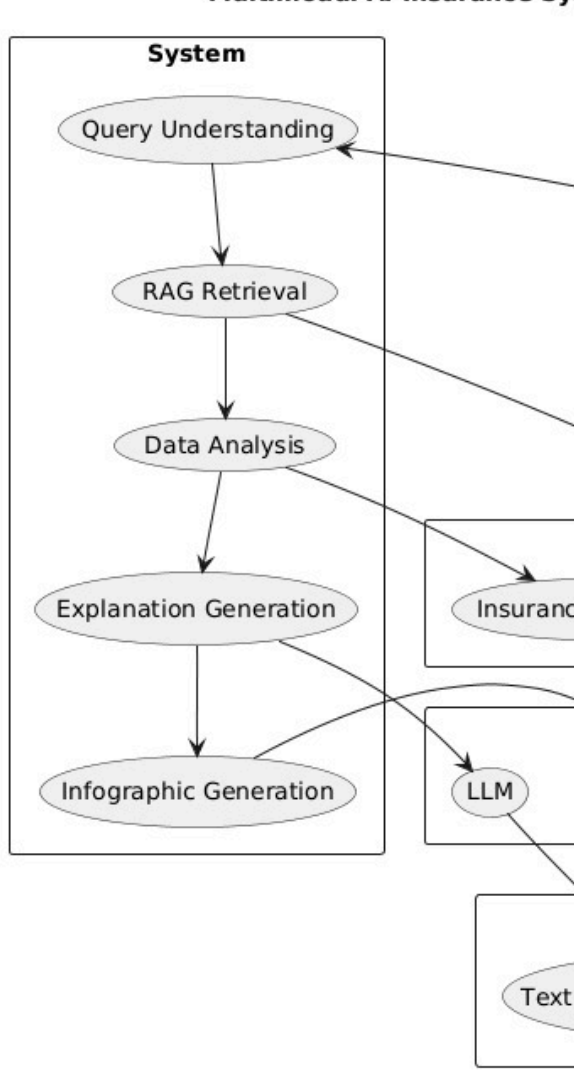
Proposed Solution -

1.Key features:

- Multimodal output (Text + Image)
- Retrieval-Augmented Generation (RAG)

- Data-driven insights from datasets
- Agent-based system (CrewAI)
- Infographic generation
- Reduced hallucination
- Context-aware responses

2.Overall Architecture / Workflow:



Explanation of Architecture Diagram

- The system starts with the user, who provides an insurance-related query through the interface.

- The query is first processed by the Query Understanding module, which identifies the intent and required output.
- The RAG Retrieval component fetches relevant information from policy documents, ensuring accurate and context-based knowledge.
- The Data Analysis module processes structured data from the insurance dataset to extract meaningful insights and patterns.
- The Explanation Generation module uses the LLM (Large Language Model) to generate clear and human-readable explanations.
- The Infographic Generation module uses an image model to convert the explanation and insights into a visual representation.
- The system combines outputs from both models to produce a multimodal result, including:
- Text explanation
- Data insights
- Infographic visualization
- Finally, the generated output is delivered back to the user in an easy-to-understand format.

3.Tools & Technologies Used :

- Programming Language: Python
- LLM APIs: Gemini
- RAG Components: Vector Database
- Datasets: Insurance datasets (FAISS)
- Agent Framework: CrewAI
- Image Generation Models: DALL·E / Stable Diffusion
- Libraries:

1.Pandas, NumPy (data analysis)

2.Scikit Learn

- Backend: Python FAST APIs

Results & Output -

1.Screenshots/Outputs

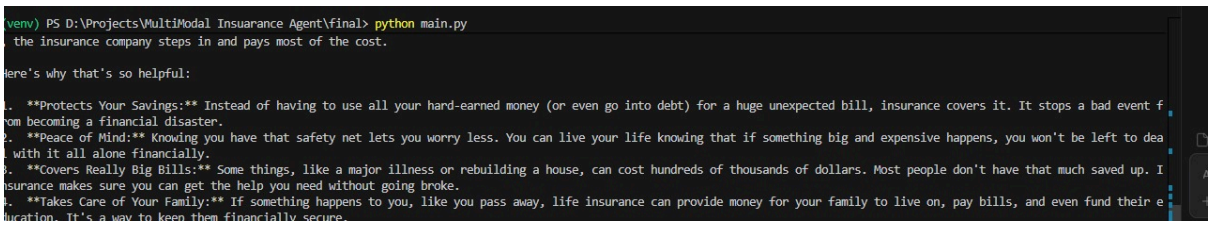


Figure 1.1(Text Output)

Figure 1.2(Image Output)

2.Reports/Dashboards/Model

The system generates multiple analytical and visual components to enhance understanding of insurance concepts. These include:

a) Analytical Reports

- Automatically generated reports based on user queries
- Summarize key insurance factors such as:
  - Driver age vs premium
  - Accident history vs risk level
- Provide data-backed explanations using structured datasets
- Include insights derived from statistical analysis

b) Interactive Dashboards

- Visual representation of insurance data using charts and graphs
- Displays relationships such as:

- Age vs Premium trends
- Risk level vs Claim probability

- Helps users easily interpret complex data
- Can be extended to real-time dashboards for business use

c) Machine Learning / AI Models

- RAG (Retrieval-Augmented Generation) Model

- Retrieves relevant information from insurance documents and datasets
- Ensures accurate and context-aware responses

- Data Analysis Models

- Perform statistical analysis on structured insurance data

- Identify patterns and trends (e.g., high-risk profiles)

- Text Generation Model (LLM)

- Generates clear and human-like explanations

- Image Generation Model

- Converts insights into professional infographic visuals

3.Key Outcomes

- Successfully generated accurate and contextual explanations
- Reduced hallucinations using RAG
- Combined structured + unstructured data effectively
- Produced professional infographic visuals
- Improved understanding of insurance concepts
- Demonstrated agent collaboration using CrewAI

Conclusion

The Multimodal Generative AI System for Insurance Concept Visualization successfully demonstrates the integration of advanced AI techniques to simplify and enhance the understanding of complex insurance concepts. By combining Retrieval-Augmented Generation (RAG), structured data analysis, and multimodal content generation, the system is able to produce accurate, context-aware, and informative outputs. The project effectively utilizes both structured datasets and unstructured policy documents to generate meaningful insights, ensuring that the responses are not only descriptive but also data-driven. The implementation of an agent-based architecture further improves system efficiency by assigning specific tasks to different agents, enabling smooth and organized workflow execution.

Moreover, the ability to generate visual infographics along with textual explanations makes the system more interactive and user-friendly, helping users better understand relationships such as risk factors and premium calculations. This enhances the overall user experience and makes the information more accessible, even for non-technical users.

In conclusion, the project achieves its goal of developing a reliable, scalable, and intelligent solution that bridges the gap between data, knowledge, and visualization, making insurance concepts easier to interpret and analyze.

Future Scope & Enhancements –

- Future Scope & Enhancements
- Integration with real-time insurance APIs
- Support for more domains beyond insurance
- Advanced interactive dashboards
- Improved infographic customization
- Voice-based query input
- Deployment as a web or mobile application
- Use of fine-tuned domain-specific LLMs
- Real-time analytics and personalization
- Integration with chatbot interfaces for customer service